

Coulomb Branches and Symmetric Pairs

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Affine Hecke Algebras

$\hat{H}_A$ is generated by

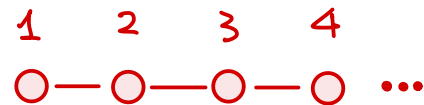
$$T_1, \dots, T_{n-1}, Y_1^{\pm 1}, \dots, Y_n^{\pm 1}$$

$$(T_i - \tau)(T_i + \tau^{-1}) = 0, \text{ for } i = 1, \dots, n-1.$$

$$T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1} \text{ for } i = 1, \dots, n-2; T_i T_j = T_j T_i \text{ if } |i-j| > 1.$$

$$T_i^{-1} Y_i T_i^{-1} = Y_{i+1}, i = 1, \dots, n-1; T_j Y_i = Y_i T_j \text{ if } j \neq i, i-1.$$

$$Y_i Y_j = Y_j Y_i \text{ for } i, j = 1, \dots, n; Y_i Y_i^{-1} = 1.$$



$\hat{H}_B$ is generated by

$$T_k, \quad X_l^{\pm 1}, \quad k = 0, 1, \dots, m-1, \quad l = 1, 2, \dots, m$$

satisfying the following defining relations:

(a) $X_l X_{l'} = X_{l'} X_l,$

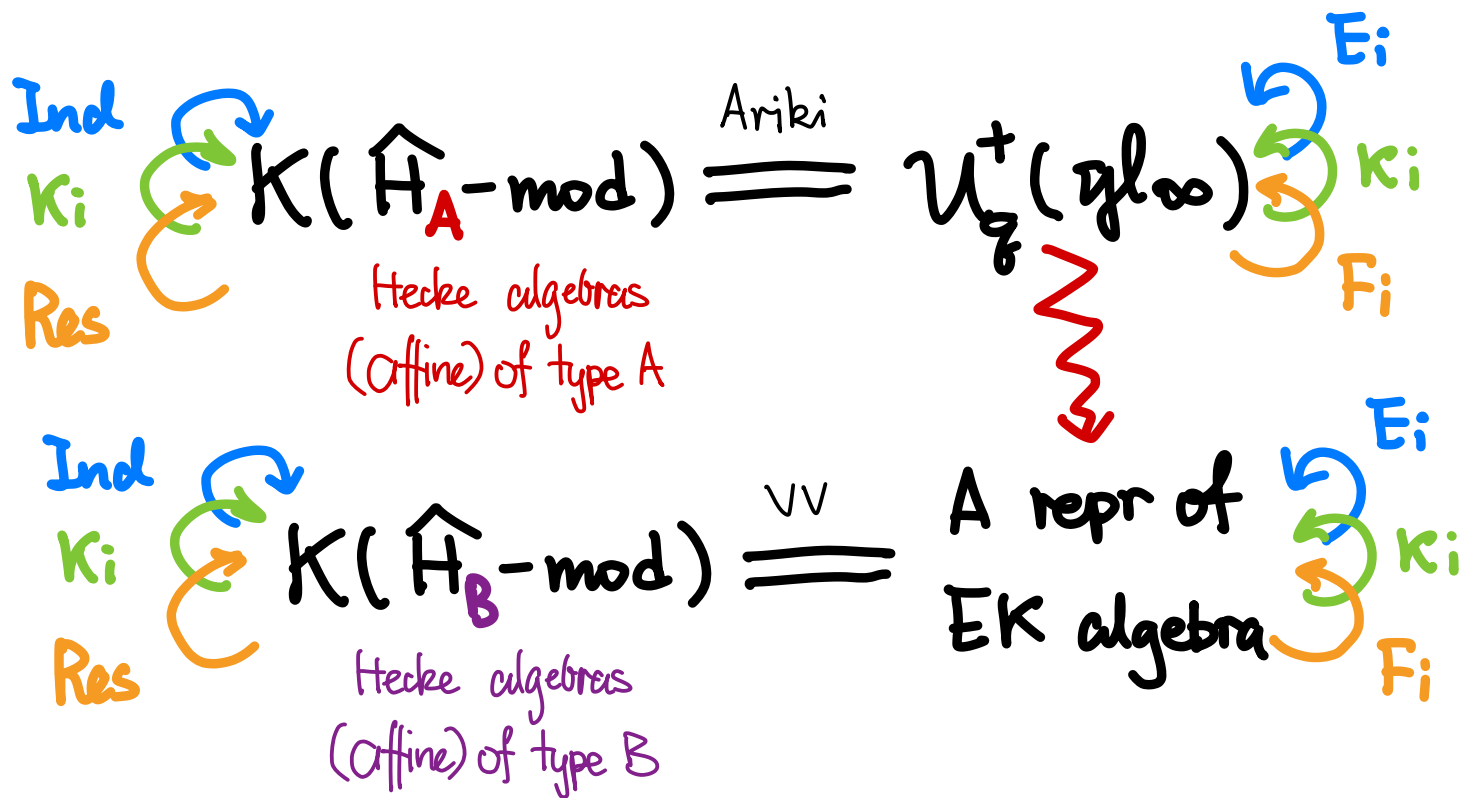
(b) $(T_0 T_1)^2 = (T_1 T_0)^2, T_k T_{k-1} T_k = T_{k-1} T_k T_{k-1}$ if $k \neq 0, 1$, and $T_k T_{k'} = T_{k'} T_k$ if $|k - k'| \neq 1,$

(c) $T_0 X_1^{-1} T_0 = X_1, T_k X_k T_k = X_{k+1}$ if $k \neq 0$, and $T_k X_l = X_l T_k$ if $l \neq k, k+1,$

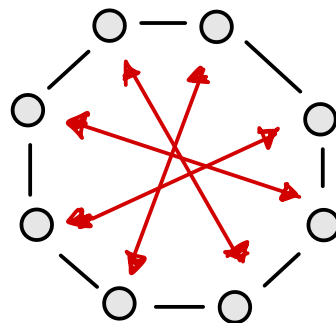
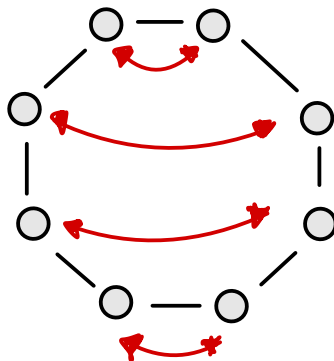
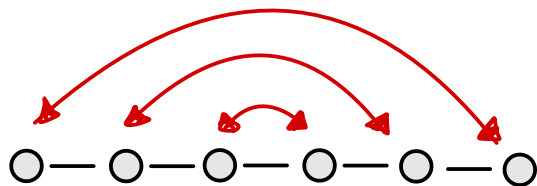
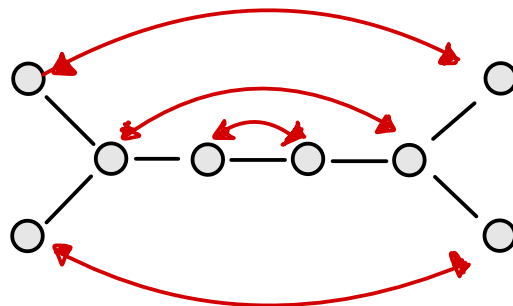
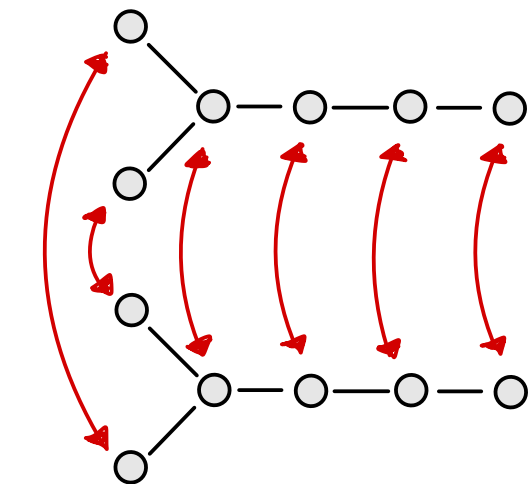
(d) $(T_k - p)(T_k + p^{-1}) = 0$ if $k \neq 0$, and $(T_0 - q)(T_0 + q^{-1}) = 0.$



Categorification



Dynkin diagram with involution



Enomoto-Kashiwara Algebras

Definition 2.1. The quantized universal enveloping algebra $U_q(\mathfrak{g})$ is the K -algebra generated by the elements e_i, f_i and invertible elements t_i ($i \in I$) with the following defining relations.

- (1) The t_i 's commute with each other.
- (2) $t_j e_i t_j^{-1} = q^{(\alpha_j, \alpha_i)} e_i$ and $t_j f_i t_j^{-1} = q^{-(\alpha_j, \alpha_i)} f_i$ for any $i, j \in I$.
- (3) $[e_i, f_j] = \delta_{ij} \frac{t_i - t_i^{-1}}{q_i - q_i^{-1}}$ for $i, j \in I$. Here $q_i := q^{(\alpha_i, \alpha_i)/2}$.
- (4) (Serre relation) For $i \neq j$,

$$\sum_{k=0}^b (-1)^k e_i^{(k)} e_j e_i^{(b-k)} = \sum_{k=0}^b (-1)^k f_i^{(k)} f_j f_i^{(b-k)} = 0.$$

Here $b = 1 - (\alpha_i^\vee, \alpha_j)$ and

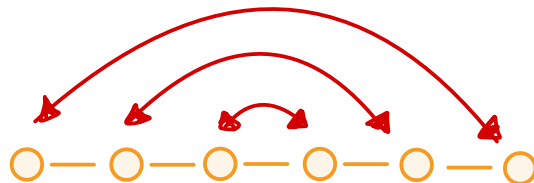
$$e_i^{(k)} = e_i^k / [k]_i!, \quad f_i^{(k)} = f_i^k / [k]_i!, \\ [k]_i = (q_i^k - q_i^{-k}) / (q_i - q_i^{-1}), \quad [k]_i! = [1]_i \cdots [k]_i.$$

$$a_{ij} = \begin{cases} 2, & i=j, \\ -1, & i \rightarrow j, \\ 0, & i \nrightarrow j. \end{cases}$$

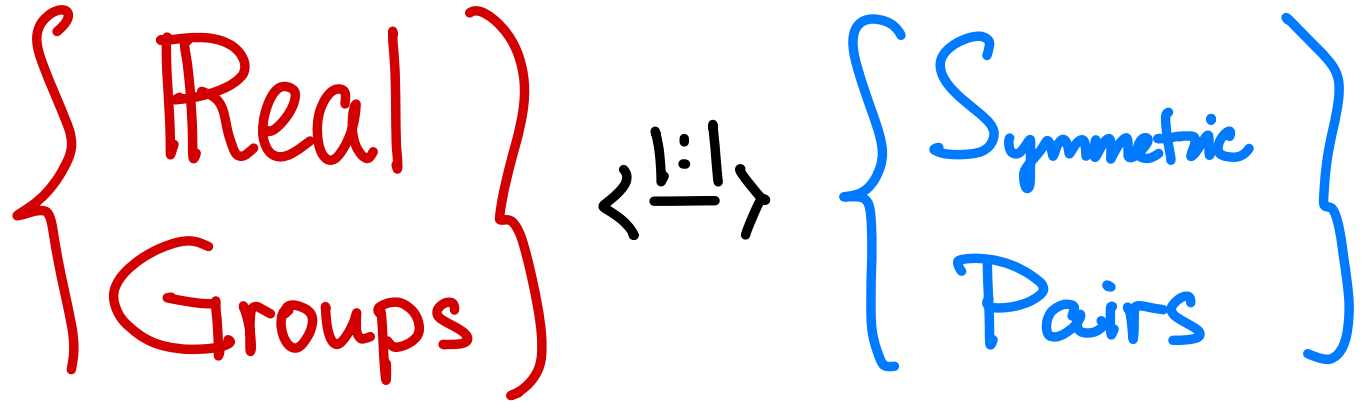


Let $\mathcal{B}_\theta(\mathfrak{g})$ be the K -algebra generated by E_i, F_i , and invertible elements T_i ($i \in I$) satisfying the following defining relations:

- $$(2.3) \quad \left\{ \begin{array}{l} \text{(i) the } T_i \text{'s commute with each other,} \\ \text{(ii) } T_{\theta(i)} = T_i \text{ for any } i, \\ \text{(iii) } T_i E_j T_i^{-1} = q^{(\alpha_i + \alpha_{\theta(i)}, \alpha_j)} E_j \text{ and } T_i F_j T_i^{-1} = q^{(\alpha_i + \alpha_{\theta(i)}, -\alpha_j)} F_j \\ \quad i, j \in I, \\ \text{(iv) } E_i F_j = q^{-(\alpha_i, \alpha_j)} F_j E_i + (\delta_{i,j} + \delta_{\theta(i),j} T_i) \text{ for } i, j \in I, \\ \text{(v) the } E_i \text{'s and the } F_i \text{'s satisfy the Serre relations.} \end{array} \right.$$



Symmetric Pairs



$$G_{\mathbb{R}} \longleftrightarrow (K = G^{\theta}, G)$$

a real form
of group G

$\theta \in G$ an
involution $\theta^2 = \text{id}$

Satake Diagrams

$g(C)$	g	t	$\dim t$	$\dim p$	$\operatorname{rk}_{\mathbb{R}} g$	Satake diagram
$\mathfrak{sl}_{l+1}(C)$ ($l \geq 1$)	$\mathfrak{sl}_{l+1}(\mathbb{R})$	$\mathfrak{so}_{l+1}$	$\frac{1}{2}l(l+1)$	$\frac{1}{2}l(l+3)$	l	
	$\mathfrak{sl}_{l+1}(\mathbb{H})$ ($l = 2p+1, p \geq 1$)	$\mathfrak{sp}_{p+1}$	$(p+1) \times (2p+3)$	$p(2p+3)$	p	
	$\mathfrak{su}_{p,l+1-p}$ ($1 \leq p \leq \frac{l}{2}$)	$\mathfrak{su}_p$ ($\oplus \mathfrak{u}_{l+1-p}$)	$p^2 + (l+1-p)^2 - 1$	$2p \times (l+1-p)$	p	
	$\mathfrak{su}_{p,p}$ ($l = 2p-1, p \geq 2$)	$\mathfrak{su}_p \oplus \mathfrak{u}_p$	$2p^2 - 1$	$2p^2$	p	
$\mathfrak{so}_{2l+1}(C)$ ($l \geq 1$)	$\mathfrak{so}_{p,2l+1-p}$ ($1 \leq p \leq l$)	$\mathfrak{so}_p$ ($\oplus \mathfrak{so}_{2l+1-p}$)	$p(2p+1) - (2l+1-p)(4l+3-2p)$	$p(2l+1-p)$	p	
$\mathfrak{sp}_{2l}(C)$	$\mathfrak{sp}_{2l}(\mathbb{R})$	$\mathfrak{u}_l$	l^2	$l(l+1)$	l	
	$\mathfrak{sp}_{p,l-p}$ ($1 \leq p \leq \frac{1}{2}(l-1)$)	$\mathfrak{sp}_p \oplus \mathfrak{sp}_{l-p}$	$p(2p+1) + (l-p)(2l-2p+1)$	$4p(l-p)$	p	
	$\mathfrak{sp}_{p,p}$ ($l = 2p$)	$\mathfrak{sp}_p \oplus \mathfrak{sp}_p$	$2p(2p+1)$	$4p^2$	p	
$\mathfrak{so}_{2l}(C)$ ($l \geq 4$)	$\mathfrak{so}_{p,2l-p}$ ($1 \leq p \leq l-2$)	$\mathfrak{so}_p \times \mathfrak{so}_{2l-p}$	$\frac{1}{2}p(p-1) + \frac{1}{2}(2l-p)(2l-p+1)$	$p(2l-p)$	p	
	$\mathfrak{so}_{l-1,l+1}$	$\mathfrak{so}_{l-1} \times \mathfrak{so}_{l+1}$	$\frac{1}{2}(l-1)(l-2) + \frac{1}{2}l(l+1)$	$l^2 - 1$	$l-1$	

$g(C)$	g	t	$\dim t$	$\dim p$	$\operatorname{rk}_{\mathbb{R}} g$	Satake diagram
$\mathfrak{so}_{2l}(C)$ ($l \geq 4$)	$\mathfrak{so}_{l,l}$	$\mathfrak{so}_l \times \mathfrak{so}_l$	$l(l-1)$	l^2	l	
	$\mathfrak{u}_{2p}^*(\mathbb{H})$ ($l = 2p$)	$\mathfrak{u}_{2p}$	$4p^2$	$2p(2p-1)$	p	
	$\mathfrak{u}_{2p+1}^*(\mathbb{H})$ ($l = 2p+1$)	$\mathfrak{u}_{2p+1}$	$(2p+1)^2$	$2p(2p+1)$	p	
E_6	E I	$\mathfrak{sp}_4$	36	42	6	
	E II	$\mathfrak{su}_2 \oplus \mathfrak{su}_8$	38	40	4	
	E III	$\mathfrak{so}_{10} \oplus \mathbb{R}$	46	32	2	
	E IV	F_4	52	26	2	
E_7	E V	$\mathfrak{su}_8$	63	70	7	
	E VI	$\mathfrak{su}_2 \oplus \mathfrak{so}_{12}$	69	64	4	
	E VII	$E_6 \oplus \mathbb{R}$	79	54	3	
E_8	E VIII	$\mathfrak{so}_{16}$	120	128	8	
	E IX	$\mathfrak{su}_2 \oplus E_7$	136	112	4	
F_4	F I	$\mathfrak{su}_2 \oplus \mathfrak{sp}_3$	24	28	4	
	F II	$\mathfrak{so}_9$	36	16	4	
G_2	G	$\mathfrak{so}_3 \oplus \mathfrak{so}_3$	6	8	2	

Quantum Groups

Definition 3.1. The quantum group $U = U_q(\mathfrak{g})$ is a $\mathbb{Q}(q)$ -algebra generated by E_i, F_i, K_i, K_i^{-1} ($i \in \mathbb{I}$), subject to the following relations, for $i, j \in \mathbb{I}$:

$$K_i K_i^{-1} = 1 = K_i^{-1} K_i, \quad K_i K_j = K_j K_i,$$

$$K_i E_j K_i^{-1} = q^{i \cdot j} E_j, \quad K_i F_j K_i^{-1} = q^{-i \cdot j} F_j,$$

$$E_i F_j - F_j E_i = \delta_{i,j} \frac{K_i - K_i^{-1}}{q_i - q_i^{-1}},$$

and, for $i \neq j$,

$$(3.1) \quad \sum_{r=0}^{1-c_{ij}} (-1)^r E_i^{(1-c_{ij}-r)} E_j E_i^{(r)} = 0, \quad \sum_{r=0}^{1-c_{ij}} (-1)^r F_i^{(1-c_{ij}-r)} F_j F_i^{(r)} = 0.$$

$$a_{ij} = \begin{cases} 2, & i=j, \\ -1, & i \rightarrow j, \\ 0, & i \nrightarrow j. \end{cases}$$

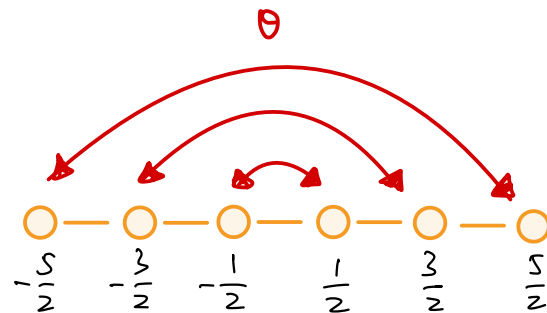


Proposition 8.5. [Let99] The $\mathbb{Q}(q)$ -algebra U^1 admits a presentation as follows:

• generators: B_i ($i \in \mathbb{I}$), $k_j^{\pm 1}$ ($j \in \mathbb{I}$);

• relations:

$$\begin{cases} k_i k_j = k_j k_i, & k_i = k_i^{-1}, \\ k_i B_j k_i^{-1} = q^{(\alpha_i - \alpha_{-i}, \alpha_j)} B_j, \\ [B_i, B_{-i}] = \frac{k_i - k_i^{-1}}{q - q^{-1}}, & \text{for } i \in \mathbb{I}^+, \\ [B_i, B_j] = 0, & \text{unless } j = -i \text{ or } |i - j| = 1, \\ B_i^2 B_j - [2] B_i B_j B_i + B_j B_i^2 = 0, & \text{for } |i - j| = 1 \text{ and } \{i, j\} \neq \{\pm \frac{1}{2}\}, \\ B_{\frac{1}{2}}^2 B_{-\frac{1}{2}} - [2] B_{\frac{1}{2}} B_{-\frac{1}{2}} B_{\frac{1}{2}} + B_{-\frac{1}{2}} B_{\frac{1}{2}}^2 = -[2] B_{\frac{1}{2}} (pq k_{\frac{1}{2}} + p^{-1} q^{-1} k_{\frac{1}{2}}^{-1}), \\ B_{-\frac{1}{2}}^2 B_{\frac{1}{2}} - [2] B_{-\frac{1}{2}} B_{\frac{1}{2}} B_{-\frac{1}{2}} + B_{\frac{1}{2}} B_{-\frac{1}{2}}^2 = -[2] (pq k_{\frac{1}{2}} + p^{-1} q^{-1} k_{\frac{1}{2}}^{-1}) B_{-\frac{1}{2}}. \end{cases}$$

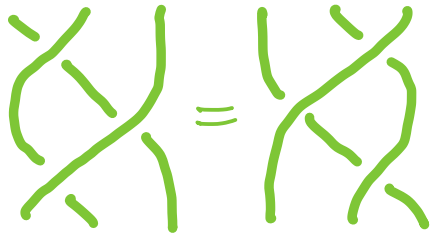


R-matrices, K-matrices

Quantum
Groups



R-matrices



Yang-Baxter
Equation



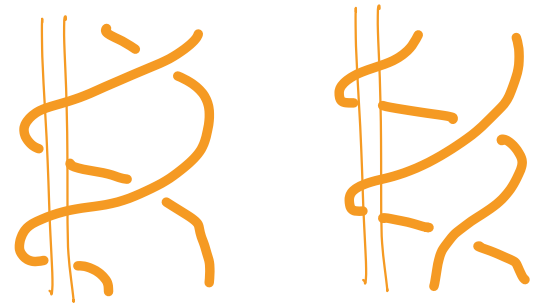
Coideal
Subalgebra

monoidal
module
category

$\mathcal{U}$ Quantum
Groups

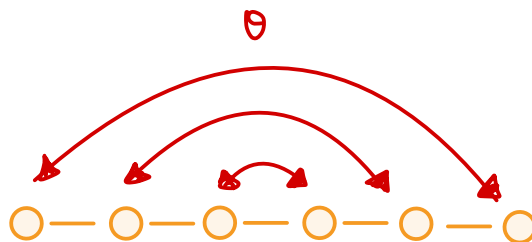


K-matrices



Reflection Equation

Wait...



Enomoto-Kashiwara
Algebra

Quantum Symmetric
Pair

$\longleftrightarrow$



oKLR

quiver

Categorification

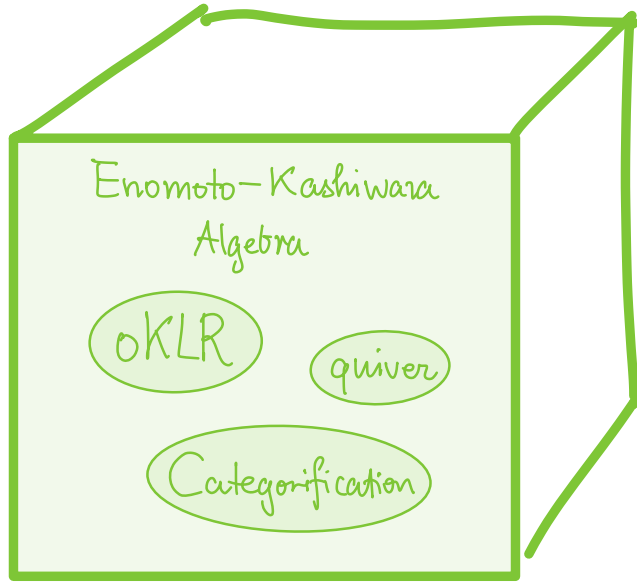
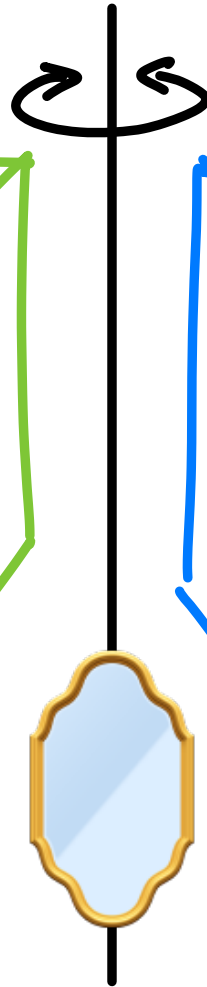
r Schur-Weyl duality

K-matrices

twisted Yangian

An answer

3d Mirror Symmetry



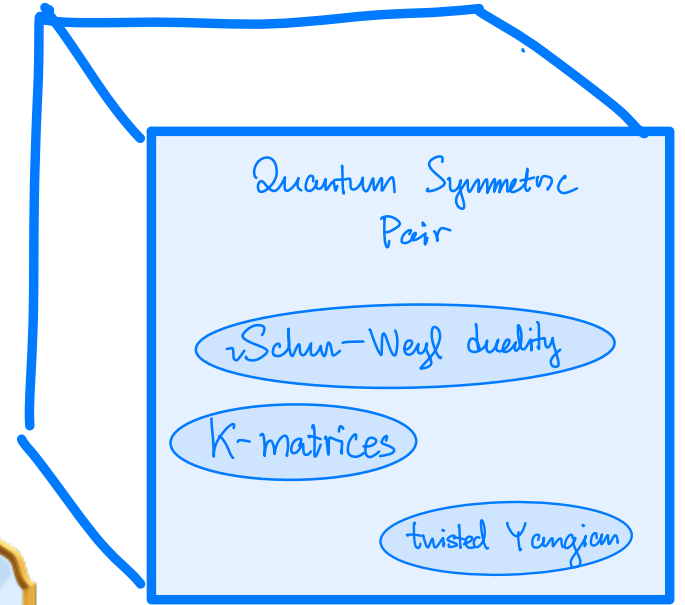
Enomoto-Kashiwara
Algebra

OKLR

quiver

Categorification

Higgs branch



Quantum Symmetric
Pair

r Schur-Weyl duality

K-matrices

twisted Yangian

Coulomb branch

Higgs vs Coulomb

a group $G \hookrightarrow N$ a representation

Higgs branch

$$\mu_{T^*N}^{-1}(0) // G$$

$$[N/G]$$

$$N //_{\theta} G$$

Coulomb branch

$$\mathrm{Spec} \left(H_*^{G^0} (R_{GN}) \right)$$

$$H_*^{G^0 \times \mathbb{C}^*} (R_{GN})$$

BFN fibers

Quiver Gauge theory

Choose a vector space V_i at each vertex i . We define

$$G = \prod_{\text{vertex } i} GL(V_i)$$



$$N = \bigoplus_{\substack{\text{arrow} \\ h: i \rightarrow j}} \text{Hom}(V_i, V_j) \\ \left(\bigoplus \text{Framing} \dots \right)$$

Quiver Gauge theory

Higgs

($N=4$)

Nakajima's
quiver
variety

" $N \geq$ "
Higgs
branch

(Lusztig's
quiver
variety)

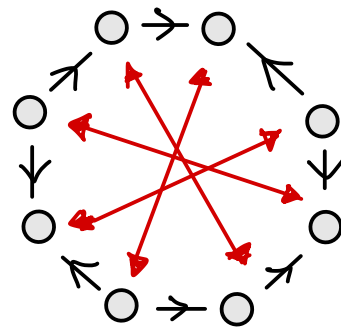
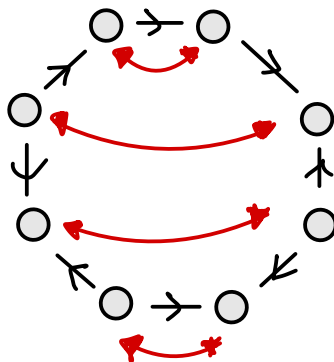
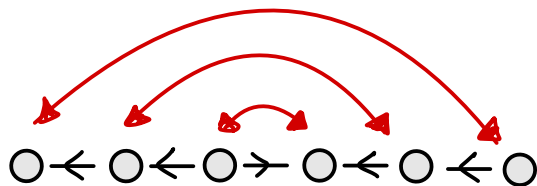
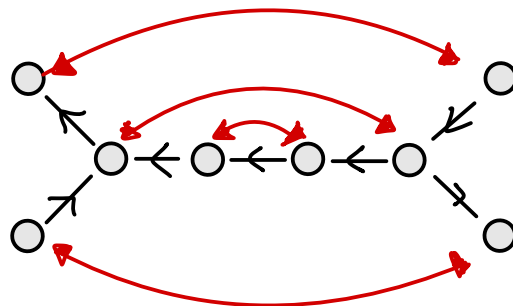
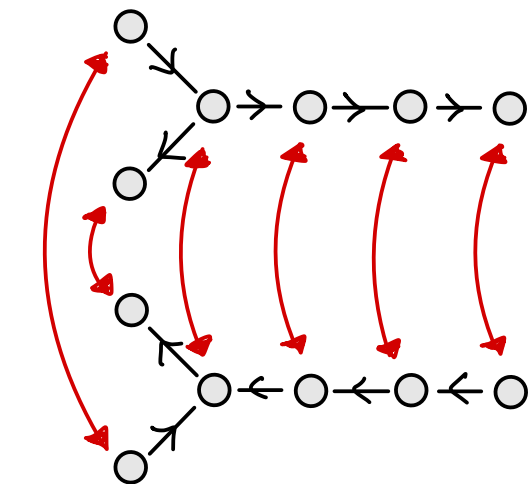
Coulomb

affine
Grassmannian
slice

(truncated
shifted
Yangian)

quantized
Coulomb
branch
algebra

Quivers with involution



i Quiver Gauge theory

Choose a symmetric bilinear form s.t. $V_i \cong V_{\tau i}^*$

$$G^T = \{g \in G: g^t = g^{-1}\} \quad (\approx \mathbf{O}(n))$$



$$N^T = \{f \in N: f^t = -f\} \quad (\approx \mathbf{O}(n))$$

($\oplus$ Framing ...)

The two sides

Higgs



Coulomb

[Li]
[Nak]

$\left(\begin{array}{c} \text{Nakajima's} \\ \text{quiver} \\ \text{variety} \end{array} \right)^{\sigma}$

$\left(\begin{array}{c} \text{affine} \\ \text{Grassmannian} \\ \text{slice} \end{array} \right)^{\theta}$

[LWW]

" $N=2$ "
Higgs
branch

$\left(\begin{array}{c} \text{Enomoto's} \\ \text{quiver} \\ \text{variety} \end{array} \right)$

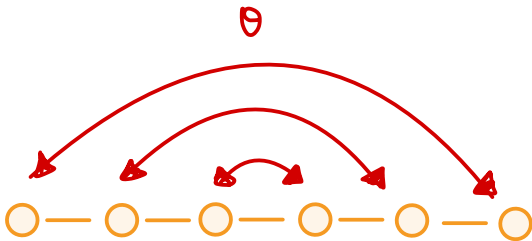
$\left(\begin{array}{c} \text{truncated} \\ \text{shifted} \\ \text{twisted} \\ \text{Yangian} \end{array} \right)$

quantized
Coulomb
branch
algebra

Yangians vs Twisted Yangians

Definition B.1. We define the “Cartan doubled Yangian” Y_∞ to be the $\mathbb{C}$ -algebra with generators $E_i^{(q)}, F_i^{(q)}, H_i^{(p)}$ for $q > 0$ and $p \in \mathbb{Z}$ and $i \in Q_0$

$$\begin{aligned} [H_i^{(p)}, H_j^{(q)}] &= 0, \\ [E_i^{(p)}, F_j^{(q)}] &= \delta_{ij} H_i^{(p+q-1)}, \\ [H_i^{(p+1)}, E_j^{(q)}] - [H_i^{(p)}, E_j^{(q+1)}] &= \frac{\alpha_i \cdot \alpha_j}{2} (H_i^{(p)} E_j^{(q)} + E_j^{(q)} H_i^{(p)}), \\ [H_i^{(p+1)}, F_j^{(q)}] - [H_i^{(p)}, F_j^{(q+1)}] &= -\frac{\alpha_i \cdot \alpha_j}{2} (H_i^{(p)} F_j^{(q)} + F_j^{(q)} H_i^{(p)}), \\ [E_i^{(p+1)}, E_j^{(q)}] - [E_i^{(p)}, E_j^{(q+1)}] &= \frac{\alpha_i \cdot \alpha_j}{2} (E_i^{(p)} E_j^{(q)} + E_j^{(q)} E_i^{(p)}), \\ [F_i^{(p+1)}, F_j^{(q)}] - [F_i^{(p)}, F_j^{(q+1)}] &= -\frac{\alpha_i \cdot \alpha_j}{2} (F_i^{(p)} F_j^{(q)} + F_j^{(q)} F_i^{(p)}), \\ i \neq j, N = 1 - \alpha_i \cdot \alpha_j \Rightarrow \text{sym}[E_i^{(p_1)}, [E_i^{(p_2)}, \dots, [E_i^{(p_N)}, E_j^{(q)}] \dots]] &= 0, \\ i \neq j, N = 1 - \alpha_i \cdot \alpha_j \Rightarrow \text{sym}[F_i^{(p_1)}, [F_i^{(p_2)}, \dots, [F_i^{(p_N)}, F_j^{(q)}] \dots]] &= 0, \end{aligned}$$



Definition 3.1. The *twisted Yangian of quasi-split type* $(\mathbb{I}, \tau)$, denoted by $\mathbf{Y}^t$, is the $\mathbb{C}[h]$ -algebra generated by $h_{i,r}, b_{i,r}, i \in \mathbb{I}^0, r \in \mathbb{N}$, subject to

$$[h_{i,r}, h_{j,s}] = 0, \quad h_{\tau i, 0} = -h_{i, 0}, \quad (3.1)$$

$$[h_{i,0}, b_{j,r}] = (c_{ij} - c_{\tau i, j}) b_{j,r}, \quad (3.2)$$

$$[h_{i,1}, b_{j,r}] = (c_{ij} + c_{\tau i, j}) b_{j,r+1} + \frac{h(c_{ij} - c_{\tau i, j})}{2} \{h_{i,0}, b_{j,r}\}, \quad (3.3)$$

$$\begin{aligned} [h_{i,r+2}, b_{j,s}] - [h_{i,r}, b_{j,s+2}] &= \frac{c_{ij} - c_{\tau i, j}}{2} h \{h_{i,r+1}, b_{j,s}\} + \frac{c_{ij} + c_{\tau i, j}}{2} h \{h_{i,r}, b_{j,s+1}\} + \frac{c_{ij} c_{\tau i, j}}{4} h^2 \{h_{i,r}, b_{j,s}\}, \end{aligned} \quad (3.4)$$

$$[b_{i,r+1}, b_{j,s}] - [b_{i,r}, b_{j,s+1}] = \frac{c_{ij}}{2} h \{b_{i,r}, b_{j,s}\} - 2\delta_{\tau i, j} (-1)^r h_{j,r+s+1}, \quad (3.5)$$

and the Serre relations: for $c_{ij} = 0$,

$$[b_{i,r}, b_{j,s}] = \delta_{\tau i, j} (-1)^r h_{j,r+s}, \quad (3.6)$$

and for $c_{ij} = -1, j \neq \tau i \neq i$,

$$\text{Sym}_{k_1, k_2} [b_{i, k_1}, [b_{i, k_2}, b_{j, r}]] = 0, \quad (3.7)$$

and for $c_{i, \tau i} = -1$,

$$\text{Sym}_{k_1, k_2} [b_{i, k_1}, [b_{i, k_2}, b_{\tau i, r}]] = \frac{4}{3} \text{Sym}_{k_1, k_2} (-1)^{k_1} \sum_{p=0}^{k_1+r} 3^{-p} [b_{i, k_2+p}, h_{\tau i, k_1+r-p}], \quad (3.9)$$

where $h_{i,s} = 0$ if $s < 0$.

Computation techniques

$$\mathcal{R}_{G,T} \longleftrightarrow \mathcal{R}_{G,T}^{\leq \omega}$$

finite codimensional
in a "trivial
bundle"

∞ -dimensional
not a vector bundle

∞ -dimensional
a vector bundle

$$G[[t]]/G[[t]] \simeq$$

$$\text{Gr}_G$$

$$\longleftrightarrow$$

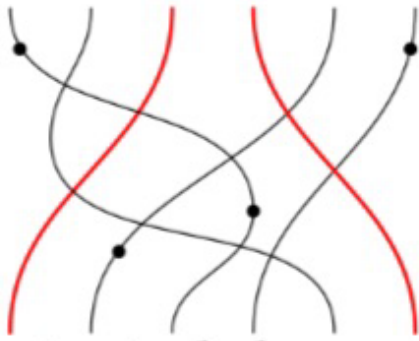
$$\overline{\text{Gr}}_G^{\omega}$$

a smooth
 G_0 -orbit

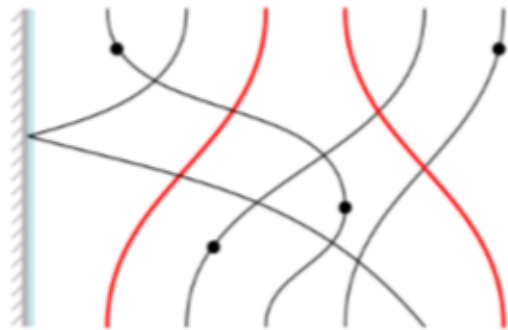
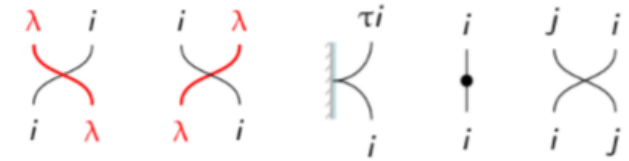
$\Rightarrow [\mathcal{R}_{G,T}^{\leq \omega}]$ can be computed (monopole)
via localization (operator)

Some diagrammatic algebras

KLRW-algebra



oKLRW-algebra



Relations with Yangians

[KTWWY]

[SSX]

module category
of $(\dots)$ KLRW

$\downarrow$

module category
of  KLRW

$\downarrow$

module category
of truncated
shifted Yangian

Higgs



duality



Coulomb

module category
of $(\dots)$ $\circ$ KLRW

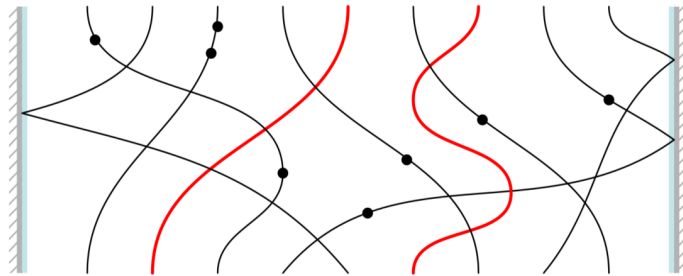
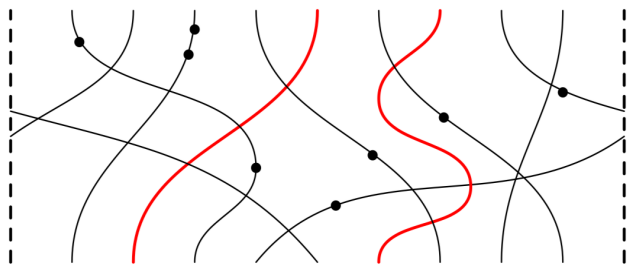
$\downarrow$

module category
of   $\circ$ KLRW

$\downarrow$

module category
of truncated
shifted twisted Yangian

Cylinder VS two-mirror



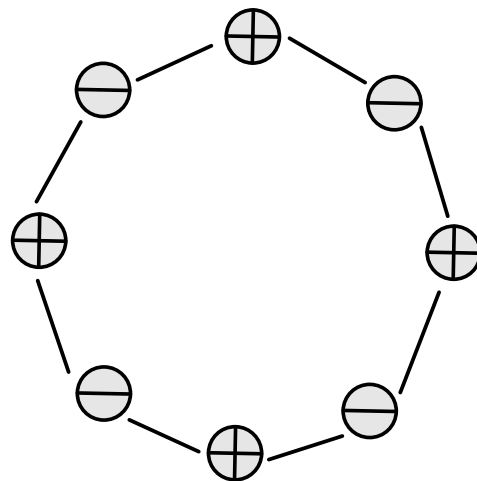
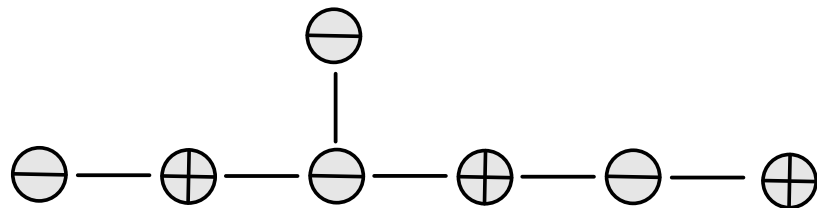
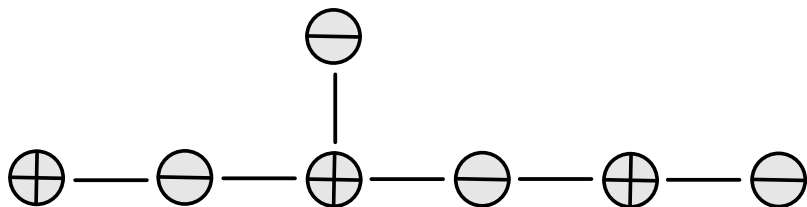
↕ Compare

DAHA of Type A

↕ Compare

DAHA of Type C

⊛ Orthosymplectic Quivers



Noncotangent type

Coulomb branch of (G, N) only depends on
 $M = N \oplus N^*$.

In general, Coulomb branch can be defined
for any symplectic representation (G, M) with
anomaly cancellation condition, i.e.

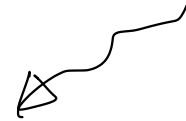
$$(\pi_4(G) \rightarrow \pi_4(Sp(M)) = \{\pm 1\}) \text{ is trivial.}$$

☆ Orthosymplectic Gauge Theory

$$G = \prod_{i \in Q_0^+} \mathrm{SO}(V_i) \times \prod_{i \in Q_0^-} \mathrm{Sp}(V_i)$$



non-cotangent type



$$M = \bigoplus_{i \rightarrow j} V_i \otimes V_j$$

($\oplus$ Framing ...)

The two sides

Higgs



Coulomb

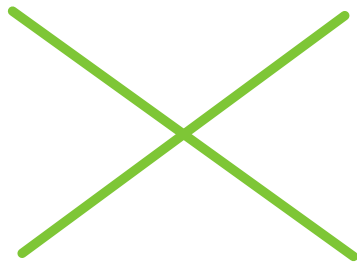
[Li]
[Nak]

(Nakajima's
quiver
variety) ^{σ}

(affine
Grassmannian
slice) ^{θ}

[LWW]

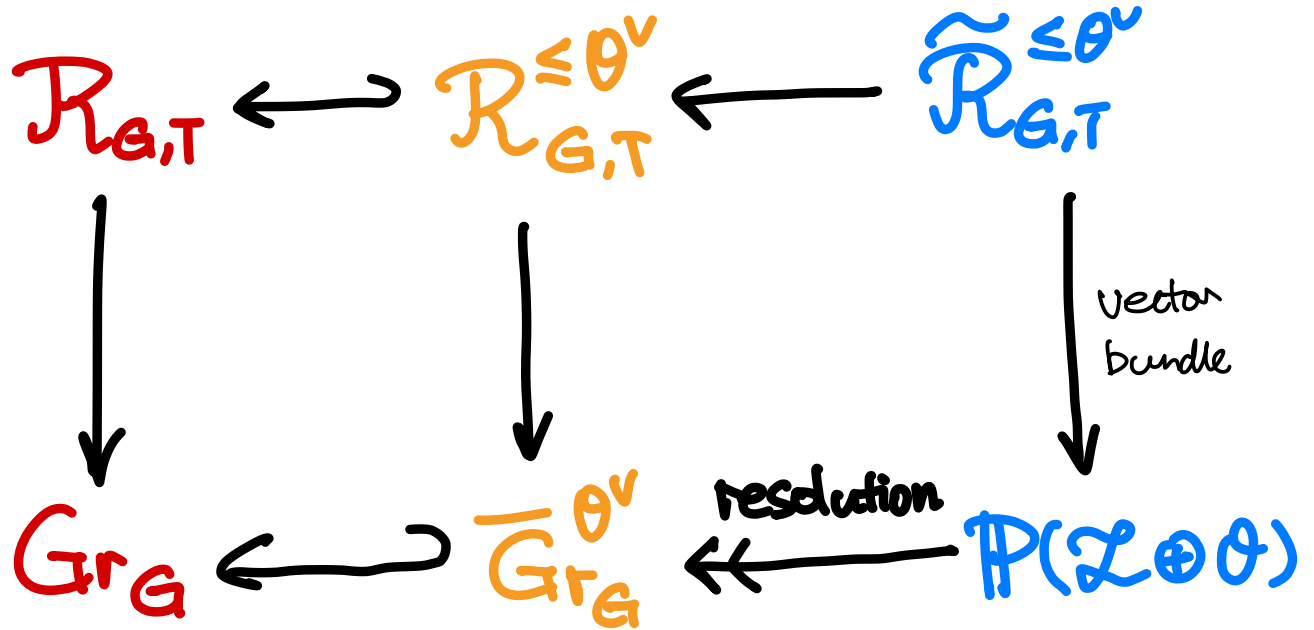
" $N=2$ "
Higgs
branch



(truncated
shifted
twisted
Yangian)

quantized
Coulomb
branch
algebra

Computation techniques



$\Rightarrow [R_{G,T}^{\leq \theta^v}]$ can be computed (monopole) via localization (operator)

Computation techniques

